

1. Problem Identification (DSM Step 1)

1.1 The Problem Statement

Animal shelters across the United States operate under significant resource constraints, including limited physical space, veterinary budgets, and staff time. Every day an animal spends in a shelter increases the operational cost and the risk of behavioral or health deterioration. The core problem is an **information gap**: shelters often lack data-driven insights into which specific animal characteristics (age, breed, health status) or external factors (listing timing) most significantly influence adoption likelihood.

1.2 Identify the Business Goal

The primary goal is to build a predictive classification model that identifies high-risk "long-stay" animals early in the intake process. By predicting the likelihood of adoption, shelter management can:

- **Allocate Resources:** Prioritize high-need animals for specialized training or medical care.
- **Target Marketing:** Focus social media and photography efforts on animals predicted to have a lower adoption probability.
- **Optimize Events:** Design adoption events based on the traits that currently show the lowest engagement.

The model will be considered successful if it identifies 70% of high-risk long-stay animals in the first 30 days and reduces average length-of-stay (LOS) for shelter animals by 10%.

1.3 Identify the Key Stakeholders

- **Shelter Operations Managers:** Need actionable insights to manage kennel capacity and staffing.
- **Veterinary Staff:** Benefit from knowing which health conditions are the biggest barriers to adoption to prioritize treatments.
- **Marketing/Outreach Teams:** Require data to justify spending on specific campaigns or platforms.
- **The Animals:** The ultimate beneficiaries of reduced length-of-stay and better-matched permanent homes.

1.4 Define the Scope of the Project

This project will focus on a **Classification Task**. I will use "AdoptionLikelihood" as the target variable (Binary: 0 for Unlikely, 1 for Likely).

- **In-Scope:** Exploratory Data Analysis (EDA) of animal traits, feature engineering (e.g., grouping age into life stages), and training/evaluating classification models (e.g., Random Forest, XGBoost).
 - **Out-of-Scope:** This phase will not include real-time deployment into a shelter software system or the analysis of image data (computer vision), though those are noted for future scope.
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2. Data Identification & Preliminary Analysis

2.1 Dataset Description

The ["Predict Pet Adoption Status" dataset](#) contains **2,000 observations** of shelter animals with the following features:

- **Categorical:** Species, Breed, Size, Color, Vaccination Status, Health Condition.
- **Numerical:** Age (months), Time in Shelter (days).
- **Target:** Adoption Likelihood (0 or 1).

2.2 Dataset Suitability

This dataset supports the problem statement because it provides a granular look at the traits that are typically available to a shelter at the time of intake. The mix of categorical and numerical data allows for the application of standard pre-processing techniques taught in the course.

The dataset presents several constraints. No behavioral data is available, which could otherwise inform LOS patterns. The dataset may also contain imbalances.

3. Preliminary Plan

1. **Wrangling:** Import the Kaggle CSV, check for missing values, and ensure data types are correct for modeling.
2. **EDA:** Visualize the correlation between "Health Condition" and "Adoption Likelihood." Identify if specific breeds or ages have skewed outcomes.
3. **Preprocessing:** Encode categorical variables and split the data into 80/20 train-test sets.
4. **Modeling:** Train a baseline Logistic Regression followed by a Random Forest Classifier to handle non-linear relationships.

5. **Evaluation:** Use a Confusion Matrix and F1-Score (to balance precision and recall) as the primary metrics, as "missing" a potential adoption is as critical as misidentifying a long-stay animal.